

How do energy forms impact energy poverty? An analysis of European degrees of urbanisation

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ABSTRACT

The access that households have to various forms of energy depends on their level of urbanisation, which leads to distinct preferences in the forms of energy they consume. Consequently, different forms of residential energy consumption should have different impacts on energy poverty, when analysed for each degree of urbanisation. Verifying this was the underlying motivation for this study. Thus, it investigates how forms of residential energy consumption (oil, wood/biomass, natural gas, and electricity) impact energy poverty in (a) cities; (b) towns and suburbs; and (c) rural areas. To do so, data from 2005 until 2018 for twelve European countries was analysed using three-panel data sets, one for each urbanisation degree. The results, estimated using Feasible Generalised Least Squares, indicated that, in sparsely populated areas (towns, suburbs, and rural areas), primary energy sources, such as wood/biomass and natural gas alleviate energy poverty. Conversely, in cities, residential electricity consumption has been the key form for decreasing both poverty and energy poverty. The results of this research highlight that energy forms have differing impacts on energy poverty in areas with different levels of urbanisation. Therefore, energy policies should be planned at a disaggregated regional level within countries, and not at a national or European level.

1. Introduction

According to the European Energy Poverty Advisory Hub (EPAH), around 34 million European citizens suffer from energy poverty (EPAH, 2022). The definition of energy poverty has been extensively discussed in the literature (Fabbri and Gaspari, 2021; Moore, 2012; Sareen et al., 2020; Thomson et al., 2017), as has the way it is measured (Sareen et al., 2020; Siksnyte-Butkiene et al., 2021; Villalobos et al., 2021). However, the contentious nature of the academic debate on the subject further demonstrates the difficulty of precisely defining energy poverty, its measurement, and the best indicators for quantifying it. Even though the need for a common definition and form of quantifying energy poverty is generally agreed (particularly in Europe), there is no consensus on the wording of this definition or the indicator to be used to measure it (Bardazzi et al., 2021). Furthermore, the widely varying information and estimations on energy poverty in the literature makes devising and developing suitable policies extremely difficult. Faced with this ambiguous range of concepts regarding energy poverty, we decided, for this study, to follow the definition and form of measurement used by both the EPAH and the United Nations. This was to enable the study to

produce accurate information and provide guidance for decision-making consistent with EPAH objectives and Sustainable Development Goals (SDGs).

This research is focused on SDG 1.2: “by 2030, reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions”. In July 2017, the General Assembly of the United Nations requested the adoption of global indicators for each SGD (Nations, 2017). As an indicator for SDG 1.2, it was suggested: (i) “Proportion of population living below the national poverty line, by sex and age”; and (ii) “Proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions” (Nations, 2017). As the primary data for evaluating energy poverty, the EPAH chose consensual, expenditure-based indicators, namely, arrears on utility bills, low absolute energy expenditure, high share of income spent on energy, and inability to keep homes adequately warm (EPAH, 2022). Therefore, in this study of energy poverty, we also focused on energy expenditure and poverty thresholds, and considered that energy poverty occurs when energy expenses account for a percentage of a household's income high enough to impact its ability to meet basic needs, such as nutrition. This definition of

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energy poverty highlights the choice some households face between spending their budget on energy consumption and costs, or on food, hygiene and other related items. To better understand this phenomenon, this paper studied the impact of various forms of residential energy consumption (oil, natural gas, wood/biomass, and electricity) on energy poverty in European countries.

The aim of EPAH, the central platform of energy poverty expertise in Europe, is to “eradicate energy poverty while accelerating energy transition”. This objective is also shared with SDG 7 - “ensure access to affordable, reliable, sustainable and modern energy for all”. The objectives of the EPAH, and of SDGs 1.2 and 7 are in line with current European policy, and were included in the Clean Energy for all Europeans package adopted in 2019 (European Commission, 2019). The Clean Energy for all Europeans package aims to move consumption away from fossil fuels toward cleaner energy sources, while protecting vulnerable consumers from energy poverty. This scope of this package is broad, and involves the energy performance of buildings, energy efficiency, the integration of renewable energy sources, and the design of the electricity market. Understanding how this objective of accelerating the energy transition while alleviating energy poverty can be achieved was the primary reason for undertaking this study. The ongoing energy transition from traditional, more polluting sources towards renewable energy sources in Europe will imply increasing dependence on electricity. This means that the residential sector will have to adjust the way it consumes energy. To more fully understand the risks of energy poverty that may arise from electrification, the study also assessed the consequences of residential electricity consumption by end-use, such as space heating, water heating, cooking, household appliances, and lighting.

Residential energy needs and habits vary between countries, but also within them. For instance, it is well-known that urban areas (cities) tend to rely on electricity, while rural areas prefer primary energy sources, such as wood/biomass and gas. The latter preference is often due to the inaccessibility of the electricity grid, the easy availability of ‘free’ wood/biomass, or simply long-standing traditional ways of living. Consequently, the compulsory switch from primary energy sources to electricity implicit in the energy transition, may increase the risk of energy poverty for some households, making it crucial to assess whether electricity can provide greater well-being for households than other, less expensive, energy forms. For example, comparing the energy costs for households in rural areas, costs will apparently be lower and well-being greater if they consume wood/biomass rather than electricity. As this may hinder the objectives for energy transition set out above, we were prompted to study the impact that residential energy consumption (or preferences) has on energy poverty in areas with different degrees of urbanisation. Accordingly, this research analysed each European country divided into three degrees of urbanisation, namely (a) cities, (b) towns and suburbs, and (c) rural areas.

The main objective of this research is to provide an accurate knowledge base for devising energy policies by which the energy transition can be implemented without exacerbating energy poverty or traditional poverty (poverty caused by the low-income or low well-being of households). Furthermore, this research also seeks to discover how the energy transition could be beneficial to households, by alleviating energy poverty. Thus, this research aims to answer the following questions: (i) does energy poverty depends on the mix of forms of residential energy consumption? (ii) what form of energy or mix should be used to alleviate energy poverty for each degree of urbanisation? (iii) could electricity be a key form of energy for a transition towards a society without energy poverty? This research breaks new ground in the literature on the subject by (a) studying the impacts of forms of residential consumption on both energy poverty and traditional poverty; (b) analysing the impact of residential electricity consumption by end-use activities on both energy poverty and traditional poverty; and (c) assessing and comparing these results by the three degrees of urbanisation. Therefore, this work fills a gap in the literature that, to the best of our knowledge, has not yet considered the influence of an area’s degree of

urbanisation in the analysis of both energy poverty and traditional poverty.

Twelve European countries were analysed by degree of urbanisation, using three data panels, one contained the data for cities in each country, another for towns and suburbs, and the last for rural areas. They were analysed using two indicators for both energy poverty and traditional poverty. The indicators were obtained from Eurostat, and are explained in the Data Subsection of the Data and Methodology Section. The time span studied runs from 2005 until 2018. The results of this research show that policies for promoting the energy transition and for reducing energy poverty, should not be devised monolithically at a national level. In fact, the results stress that forms of residential consumption impact energy poverty in a variety of way, and depend on an area’s degree of urbanisation. In rural areas, as well as in towns and suburbs, primary energy sources (wood/biomass and natural gas) have been key to alleviating both energy poverty and traditional poverty. In contrast, in cities, residential electricity consumption has been the main way to decrease both energy poverty and traditional poverty. In addition, this research revealed that the energy-inefficiency of electric water heating systems and household appliances can limit the potential benefits electricity brings for households. Therefore, the current transition towards a green energy mix, and the ongoing electrification of economies, should be carefully planned so as not further increase energy poverty.

The rest of the paper is organised as follow. Section 2 briefly summarises the literature on energy poverty, and identifies the gap in it that this paper fills. Section 3 presents the data and describes the methodology used to analyse it. The results are set out in Section 4 and discussed in Section 5. Lastly, Section 6 contains the conclusions.

2. Literature review

The term energy poverty has several alternative labels, namely fuel poverty, energy vulnerability, energy deprivation, energy precariousness, and energy burden. All of these terms are applied in the literature depending on their context and subject (Fabbri and Gaspari, 2021). Energy poverty is an increasingly urgent problem, and has led to frequent discussions in the literature over the last few years (Bednar and Reames, 2020; Frank et al., 2006; Mould and Baker, 2017; Wang et al., 2021). The definition of energy poverty generally implies that a household has to pay a disproportionate cost for energy use (Bednar and Reames, 2020; Wang et al., 2021). The literature argues that this disproportionate cost may be caused by high energy prices, low household income, or inefficient energy use (Bednar and Reames, 2020; Frank et al., 2006; Mould and Baker, 2017; Wang et al., 2021). Energy poverty can also imply that a household cannot afford to consume the energy needed to keep their home adequately warm (Fabbri and Gaspari, 2021; Gouveia et al., 2018). This definition is often used in the European context.

The definition of energy poverty has been twofold and, to measure it, a wide range of affordability indicators have been agreed on (Wang et al., 2021). These affordability indicators measure energy poverty by analysing the percentage of a household’s total income used for energy. In 1991, the first definition of energy poverty was coined, along with the first measurement to quantify it – the 10% indicator (Boardman, 1991). The definition described energy poverty as the condition in which a household spend more than 10% of its income on energy services (Boardman, 1991). The 10% indicator was further defined as the heating-related energy expenditure required to maintain an adequate level of warmth in homes (Moore, 2012). Criticism of the 10% indicator led to the United Kingdom (UK) Fuel Poverty strategy (in which the first energy poverty indicators were introduced), and three other energy poverty definitions and indicators were developed (Hills, 2011; Hills, 2012; Moore, 2012). However, these indicators also received criticism, mainly for reducing the number of households classified as energy-poor, and for aggregating the indicators of fuel poverty and energy efficiency

(Bednar and Reames, 2020; Wang et al., 2021). Multi-index energy poverty measures have been explored by the literature (Siksnelyte-Butkiene et al., 2021; Villalobos et al., 2021). These energy poverty definitions also consider aspects such as the inefficiency of buildings and appliances, energy affordability, public energy-related services, and the quality and reliability of national energy systems (Siksnelyte-Butkiene et al., 2021; Villalobos et al., 2021).

The literature often argues that energy poverty has been caused by an economic crisis, unemployment, low income, and the poor energy-efficiency of housing (Fabbri and Gaspari, 2021; Gouveia et al., 2018; Siksnelyte-Butkiene et al., 2021). Economic assistance, in the form of subsidies, has been the main policy of European countries to minimise the burden of energy costs for households with low incomes or threatened by energy poverty (Santamouris, 2016). However, this solution does not provide energy-poor households with a long-term solution. On the contrary, it merely minimises the risk of poverty in the short term (Gouveia et al., 2018; Santamouris, 2016). This subsidy policy does not generate consistent and sustainable savings for low-income households, that would allow them to improve the energy-efficiency of their homes. In studies on this aspect, the quality of insulation materials, window frames, and household appliances are taken into consideration, and it is argued that the cost of increasing the energy-efficiency of homes is likely to be too high for households already in need (Gouveia et al., 2018; Santamouris, 2016). Therefore the literature argues that policymakers should promote rehabilitation programmes or subsidies, to improve the energy performance of buildings, thereby providing a long-term solution for households threatened by energy poverty and poverty (Koomson and Danquah, 2021).

From the extensive literature on this subject, it can be concluded that energy poverty occurs when lack of income forces individuals to choose between energy consumption and basic needs, such as food and hygiene. The Covid-19 pandemic and resulting government emergency measures like lockdowns, are expected to aggravate energy poverty worldwide (Bardazzi et al., 2021; Bienvenido-Huertas, 2021). In fact, lockdowns have provoked an increase in residential energy consumption as families spend more time at home. Consumption has also increased due to the rise of the home office and of e-learning by children. Given all these changes, it is more important than ever to gain a more detailed understanding of patterns of domestic energy consumption and how they might impact energy poverty, something the literature has not yet done. Consequently, this research into energy poverty in Europe also analyses the various forms of residential energy and its end-use.

The literature suggests that households in areas with different degrees of urbanisation may have dissimilar energy needs, with differing impacts on energy poverty. This divergence was identified in research on the relationship between population density and energy use (Glaeser and Kahn, 2010; Larson and Yezer, 2015). However, this literature did not specifically look at energy poverty (Mahumane and Mulder, 2022). Conversely the literature on energy poverty and its various aspects, has not yet considered the role of urbanisation or the impact of population density on energy poverty (Mahumane and Mulder, 2022; Pelz et al., 2018). Consequently, the impact of urbanisation on energy poverty is still underexplored (Mahumane and Mulder, 2022). Therefore, this research also aims to fill this gap in the literature by taking into account the levels of urbanisation within European countries and analysing its respective impact on energy poverty.

3. Methodology

Three panels of data were used to study the impact of various forms of residential energy and residential electricity consumption by end use on both energy poverty and traditional poverty, and were analysed by three degrees of urbanisation: (a) Cities; (b) Towns and Suburbs; and (c) Rural Areas. The initial intention was to study all European Union countries, but the absence of national observations for the core dependent variables led to the exclusion of 15 countries. The criterion used

meant that countries with data on less than 50% of the temporal sample in two or more dependent variables were excluded. This criterion was employed because panel econometrics best practice suggest that, for a viable comparison from which conclusions can be drawn, panel models should be estimated using the same countries and timeframe. Therefore, only twelve European countries were analysed, namely: Austria, Belgium, France, Greece, Ireland, Italy, the Netherlands, Poland, Portugal, Spain, Sweden, and the United Kingdom over a period from 2005 until 2018. This section describes how the characteristics of the data and model specification tests were evaluated, and how the models and estimators were determined, but begins by indicating the data used and assessed.

3.1. Data

The Eurostat tables computed by the European Union on Income and Living Conditions (EU-SILC) database were used to obtain the indexes of both energy poverty and traditional poverty. The data obtained was already disaggregated by degree of urbanisation from the Eurostat database. Table 1 shows the name of the Eurostat table and respective index and table code of the variables retrieved. It should be noted that all variables are measured annually. The research undertaken here focused on two indicators of energy poverty and two indicators of traditional poverty, and are described in detail below. The suffixes C, TS, and RA represent the three degrees of urbanisation: Cities, Towns and Suburbs, and Rural Areas, respectively. The prefixes L and D denote the natural logarithms and first differences of the variables. Table 2 contains the descriptive statistics of the raw data, while Fig. 1 shows in graphical form, the indexes of energy poverty and traditional poverty by degree of urbanisation.

The *at-risk-of-poverty rate* and *people at risk of poverty and social exclusion* were the indicators used to measure traditional poverty as indicated by the literature (Brunner et al., 2012; Karpinska and Śmiech, 2020b). *At-risk-of-poverty* describes people with an income below the risk of poverty threshold, specifically, households with an equivalised income after social transfers below 60% of the national income (European standard of relative income poverty threshold). *People at risk of poverty and social exclusion* are those families or individuals who (a) are at risk of poverty; (b) have worked less than 20% of their potential during the previous 12 months (*People living with very low work intensity*); or (c) are severely materially deprived. The *Severely materially deprived* are those households/individuals who are unable to afford at least four

Table 1
Eurostat data details.

Table name	Index	Table code
<i>At-risk-of-poverty rate by degree of urbanisation^a</i>	Annual percentage	ilc_li43
<i>People at risk of poverty or social exclusion by degree of urbanisation^b</i>	Annual percentage	sdg_01_10a
<i>At-risk-of-poverty rate after deducting housing costs by degree of urbanisation^c</i>	Annual Percentage	ilc_li48
<i>Housing cost overburden rate by degree of urbanisation^d</i>	Annual Percentage	tessi165
<i>Average number of rooms per person by degree of urbanisation^e</i>	Annual average	ilc_lvho04d
<i>Mean and median income by degree of urbanisation^f</i>	Annual Euro	ilc_di17
<i>People living in households with very low work intensity^g</i>	Annual number of households	tipslc40

^a https://ec.europa.eu/eurostat/web/products-datasets/-/ilc_li43.

^b https://ec.europa.eu/eurostat/web/products-datasets/-/sdg_01_10a.

^c https://ec.europa.eu/eurostat/web/products-datasets/-/ilc_li48.

^d <https://ec.europa.eu/eurostat/web/products-datasets/-/tessi165>.

^e https://ec.europa.eu/eurostat/web/products-datasets/-/ilc_lvho04d.

^f https://ec.europa.eu/eurostat/web/products-datasets/-/ilc_di17.

^g <https://ec.europa.eu/eurostat/web/products-datasets/-/tipslc40>.

Table 2
Descriptive statistics of urbanisation degrees variables.

Variable	Degree of urbanisation	Mean	SD.	Min	Max
Poverty					
At-risk-of-poverty rate (ARP)					
ARP-C	Cities	15.90655	3.552534	9.6	25.5
ARP-TS	Towns and suburbs	15.42024	3.933513	8.3	24.5
ARP-RA	Rural areas	18.56071	5.882095	5.2	28.2
People at risk of poverty or social exclusion (PARPSE)					
PARPSE-C	Cities	23.32679	4.582059	13.4	38.6
PARPSE-TS	Towns and suburbs	21.83869	6.519009	12.4	47.2
PARPSE-RA	Rural areas	24.36548	8.085583	6.5	50.6
Energy Poverty					
At-risk-of-poverty rate after deducting housing costs (ARP_DHC)					
ARP_DHC-C	Cities	31.45833	6.050619	19.7	47.8
ARP_DHC-TS	Towns and suburbs	30.10595	4.928321	19.4	48.3
ARP_DHC-RA	Rural areas	33.00893	7.216273	19.4	53.1
Housing cost overburden rate					
HCO-C	Cities	12.85119	7.870281	3.3	49
HCO-TS	Towns and suburbs	9.9	7.968689	2.5	48.1
HCO-RA	Rural areas	8.752381	7.338145	1.6	41.2
Control variables					
Average number of rooms per person (RPP)					
RPP-C	Cities	1.619643	.3323404	1	2.2
RPP-TS	Towns and suburbs	1.709524	.3452393	1	2.3
RPP-RA	Rural areas	1.800595	.3952559	1	2.7
Mean income (INC)					
INC-C	Cities	18827.79	4194.454	6864	26379
INC-TS	Towns and suburbs	17721.35	4606.879	5431	26267
INC-RA	Rural areas	16566.9	5016.091	4769	25691
People living in households with very low work intensity (HLWI)					
HLWI-C	Cities	1154.982	1130.105	87	4511
HLWI-TS	Towns and suburbs	569.8929	545.5836	1	2485
HLWI-RA	Rural areas	472.9524	433.1783	5	2227

Notes: The observations number is 168.

of the following items: (i) mortgage or rent; (ii) utility bills (energy poverty indicator); (iii) heating to keep the home adequately warm (the most studied energy poverty indicator in Europe); (iv) unexpected expenses; (v) regular consumption of meat or protein; (vi) holidays; (vii) television set; (viii) washing machines (energy poverty indicator); (ix) car; and (x) telephone. Although the *people-at-risk-of-poverty rate* and the *social-exclusion-variable rate* are traditional poverty measures, it is indeed a subjective index that contains indicators of energy poverty, as noted. It should be highlighted that *People at risk of poverty and social exclusion* is the main indicator for monitoring the European 2020 Strategy poverty target and the EU 2030 target on energy poverty and social exclusion (Eurostat,).

In this study, the indicators used to measure energy poverty were the *at-risk-of-poverty after deduction of housing-costs rate* and the *housing-costs-overburden rate*. The *at-risk-of-poverty rate after deduction of housing-costs* shows the percentage of families or individuals whose disposable income after deducting housing costs is below the standard European relative income poverty threshold (60% of median national income).

Housing-costs are those linked to the right to live in accommodation (rents or loans), including the costs of utilities (energy expenses), namely the costs of electricity, gas, and heating consumption. The *Housing-costs overburden rate* shows the percentage of families or individuals who spend more than 40% of their disposable income on housing costs. In this indicator, disposable income includes all earnings, wealth, social, and insurance transfers, and also accounts for transfers by public intervention, such as subsidies to aid households to meet their housing costs. In addition to helping evaluate the benefits of energy subsidies, the *Housing-costs overburden rate* is also comparable to the ten per cent rule index. However, as it includes other housing costs, such as rent and loans, the threshold is raised to 40%. Both these energy poverty indicators reveal how excessive consumption by inefficient electric water heating can increase energy expenses, and reduce disposal income for other crucial budget items, such as nutrition. Households or individuals are likely to face a trade-off between supporting the increase in electricity expenditures and spending their budget in food and hygiene, or other related items. In addition to being the main indicators of energy poverty used to monitor progress toward the European objectives for social protection and social inclusion (Committee, 2015), these two indicators are also used by academic literature on the subject (Deller et al., 2021; Karpinska and Śmiech, 2020a, b).

Three control variables were used, namely, the *average number of rooms per person*, *mean income*, and *people living with very low work intensity*. The control variables were also obtained broken down by degree of urbanisation. The *average number of rooms per person* reveals the number of rooms per individual in a household, as the size of dwellings is often considered an indication of wealth and an important factor in the risk of poverty and social exclusion (Thomson, 2018). *People living with very low work intensity* describes households with individuals working less than 20% of their work potential. *Mean income* is the mean income within a country by degree of urbanisation. According to the classification of Villalobos et al. (2021) both income and percentage of people living in a household with low work intensity have a first-order impact. It was expected that lower income would raise a household's likelihood of suffering from energy poverty, and higher income would lower it. The relationship between individual work intensity and energy poverty, is linked to the impact of individual productivity. Individuals with low productivity will have lower earnings from labour (wages), and their households will have less disposable income to meet energy expenses. Therefore, the greater the number of people in a household living with very low work intensity, the greater the risk of both traditional poverty, and energy poverty.

The analysis of Table 2 shows that densely populated areas, i.e., urban zones, have the highest share of households in both energy and traditional poverty. However, poverty and social exclusion is also a widespread phenomenon in rural areas, i.e., thinly populated areas throughout the European Union. In fact, the highest maximum values for both energy and traditional poverty indicators are found in rural areas.

Fig. 1 shows that for all three degrees of urbanisation, the indicator *At-risk-of-poverty rate after deducting housing costs (ARP_DHC)* is high in all countries. This shows that energy poverty is a society-wide problem, regardless of geographic location. Moreover, the more subjective indicator for traditional poverty measure used in this research, the *PARPSE*, closely follows the *ARP_DHC*, in terms of magnitude and trend. This further highlights the prevalence of household poverty as a consequence of energy prices/consumption. The percentage of households spending more than 40% of their income on housing costs, by all degrees of urbanisation, was relatively low in all countries (except Greece). In contrast, the other indicators of poverty measures had higher percentages, suggesting that both energy and traditional poverty may be more related to income than to energy consumption habits. This is one of the areas this study sought to scrutinise.

To the best of our knowledge, the impact on both energy and traditional poverty of forms of residential consumption of energy and of electricity by end-use is unexplored. To discover their impact the study

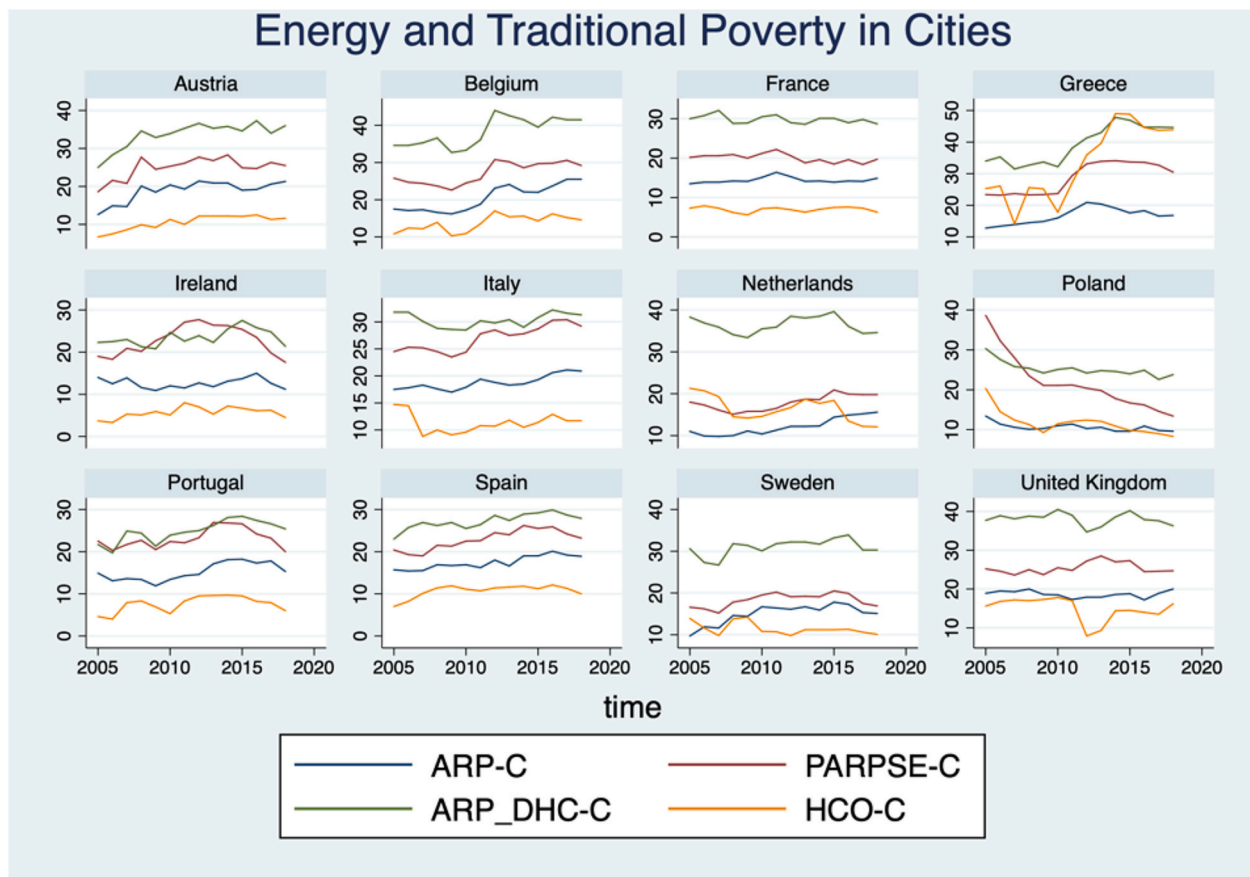


Fig. 1. Energy and Traditional Poverty by urbanisation degree.

tested various forms of residential energy consumption, namely, oil, natural gas, wood/biomass, and electricity, and various end-uses of residential electricity consumption, such as space heating, water heating, cooking, electrical appliances and lighting. These variables were retrieved from the ODYSSEE-MURE database (ODYSSEE-MURE, 2021). The descriptive statistics of the raw data for the energy related variables are presented in Table 3. The energy-related variables measure annual residential consumption in million tonnes of oil equivalent. It should be noted that the energy-related variables and the indicators of energy poverty and traditional poverty are all measured annually. Residential coal consumption in Europe is relatively inconsequential, as shown by (Thonipara et al., 2019) and by Table S2 in the supplementary material, registering around 0% in all European countries except the Czech Republic and Poland. Consequently, in this research, residential coal consumption was not considered.

Table 3 reveals that natural gas is the main form of residential consumption, followed by electricity. Natural gas can be used for space and water heating, and for cooking, whereas electricity can be used for both these activities and also to power household appliances and lighting, which, as Table 3 shows, consume the most residential electricity.

3.2. Cross-sectional dependence and unit roots tests

As this research intended to analyse energy poverty and traditional poverty in European countries, common shocks and spatial dependence were expected. Common shocks could be provoked by the common European policies and guidance on eradicating social exclusion and poverty, including energy poverty. Furthermore, European levels of urbanisation share certain features, such as population density and forms of access to energy, which may also cause spatial dependence.

Therefore, it was highly advisable to check for the presence of cross-sectional dependence in the data (Pesaran, 2004). A battery of cross-sectional dependence tests were carried out, including the Breusch-Pagan LM and Pesaran CD tests, which are two of the most frequently used in the literature (Abban and Hasan, 2021; Dogru and Bulut, 2018). The four cross-sectional dependence tests, (i) the Breusch-Pagan LM (Breusch and Pagan, 1980); (ii) the Pesaran Scaled LM (Pesaran et al., 2008); (iii) the Bias-corrected scaled LM (Baltagi et al., 2012); and (iv) the Pesaran CD (Pesaran, 2004), were performed on variables in both levels and first differences. Table S1, in the supplementary material, shows the cross-sectional dependence tests.

The data used in this research was for a fixed number of time periods ($T = 13$) and cross-sections ($N = 12$), which meant the panel were neither small nor large (Baltagi et al., 2012; Pesaran et al., 2008). The Breusch-Pagan LM is robust when both T and N are small, and the remaining cross-sectional dependence tests are valid for large N and T . The results indicated the presence of cross-sectional dependence in all variables at levels and in most of the first-difference variables. Consequently, the integration order of the series was assessed using the CIPS second generation unit roots test proposed by (Pesaran, 2007). The results are shown in Table 4.

As Table 4 shows, the CIPS test indicated that most of the variables are integrated of order one, $I(1)$, and the others are $I(0)$. Therefore, all the estimations are performed with the variables at first differences.

3.3. Models

To accomplish one of the main objectives of this work, two groups of models were performed to analyse traditional poverty (*Traditional Poverty models*) and energy poverty (*Energy Poverty models*) and were

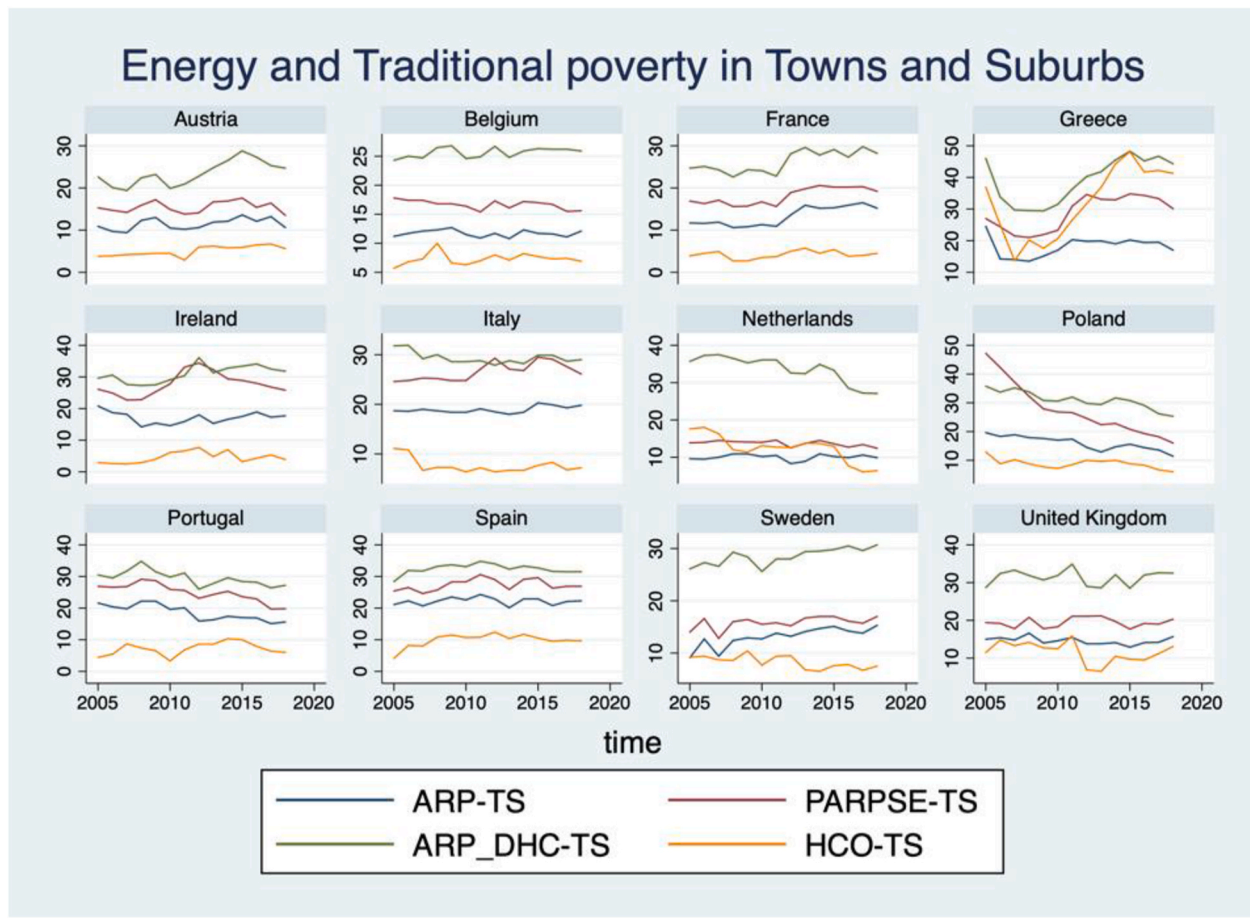


Fig. 1. (continued).

estimated for each degree of urbanisation. The variables in the *Traditional Poverty models* are *at-risk-of-poverty rate* (ARP) and *people at risk of poverty and social exclusion* (PARPSE). While in the *Energy Poverty models*, the dependent variables are *at-risk-of-poverty after deducting housing costs rate* (ARP_DHC) and *housing-costs overburden* (HCO). The correlation matrix revealed a strong correlation between the variables for forms of residential energy consumption and the variables for residential electricity consumption by end-use. Additionally, the variance inflation factors indicated that the models should not be estimated using all the energy-related variables, because of multicollinearity issues. Consequently, the models were divided into two subgroups, namely *End-use of electricity models* and *Forms of energy models*. Table 5 summarises the groups and subgroups of models and presents their nomenclature to aid in their identification.

The possible presence of the collinearity and multicollinearity was ruled out after energy-related variables were divided into different models. The highest correlation value recorded was 0.65, and the highest variance inflation factor was 3.7.

3.4. Selection of specification tests and estimators for the models

It is always advisable to analyse a model's characteristics before it is estimated. Therefore, a battery of tests were conducted to evaluate the models' characteristics, particularly, to determine if they are appropriately specified (see Table 6). The Ramsey Reset test (Ramsey, 1969) indicated that the models had no omitted variable problems. The Modified Wald test for groupwise heteroskedasticity (Baum, 2001) detected the presence of heteroskedasticity in all models except two (C-II and C-VI). Regarding serial correlation, the Wooldridge test (Wooldridge, 2010) only revealed its existence for model RA-I. To detect the

phenomenon of contemporaneous correlation, four distinct tests were performed; (i) the Breusch-Pagan LM test (Greene, 2003), (ii) the Pesaran's test (Pesaran, 2004); (iii) the Frees test (Frees, 1995); and (iv) the Friedman test (Friedman, 1937). The results were not consensual, and contemporaneous correlation was detected in only a few models. Lastly, the Hausman test suggested the presence of random effects in all models.

The Panel Corrected Standard Error (PCSE) and the Feasible Generalised Least Squares (FGLS) estimators are both suitable for dealing with random effects panel data. The PCSE estimator is more suited to analysing finite samples than the asymptotically efficient FGLS (Chong et al., 2019; Reed and Ye, 2011). However, the FGLS estimator is more efficient and robust for analysing panel data with a larger number of time observations than cross-sections (Beck and Katz, 1995; Reed and Ye, 2011). Furthermore, the FGLS is more efficient for evaluating data on first differences, i.e., first differentiated models (Beck and Katz, 1995; Reed and Ye, 2011). The characteristics of the data, namely the larger number of time observations than cross-sections, as well as the integration order of the variables which led to the use of first differentiated variables, suggested that the FGLS was the most suitable estimator. Furthermore, the FGLS can still estimate models with heteroskedastic and correlated-error structures, a phenomenon highlighted in the models in the specification tests.

4. Results

The results of the analysis of the impact of forms of residential energy consumption on energy poverty by degree of urbanisation are set out in Table 7 and summarised in Fig. 2. The results of the analysis of the impact of residential electricity consumption by end-use on energy poverty by degree of urbanisation are shown in Table 8 and graphically

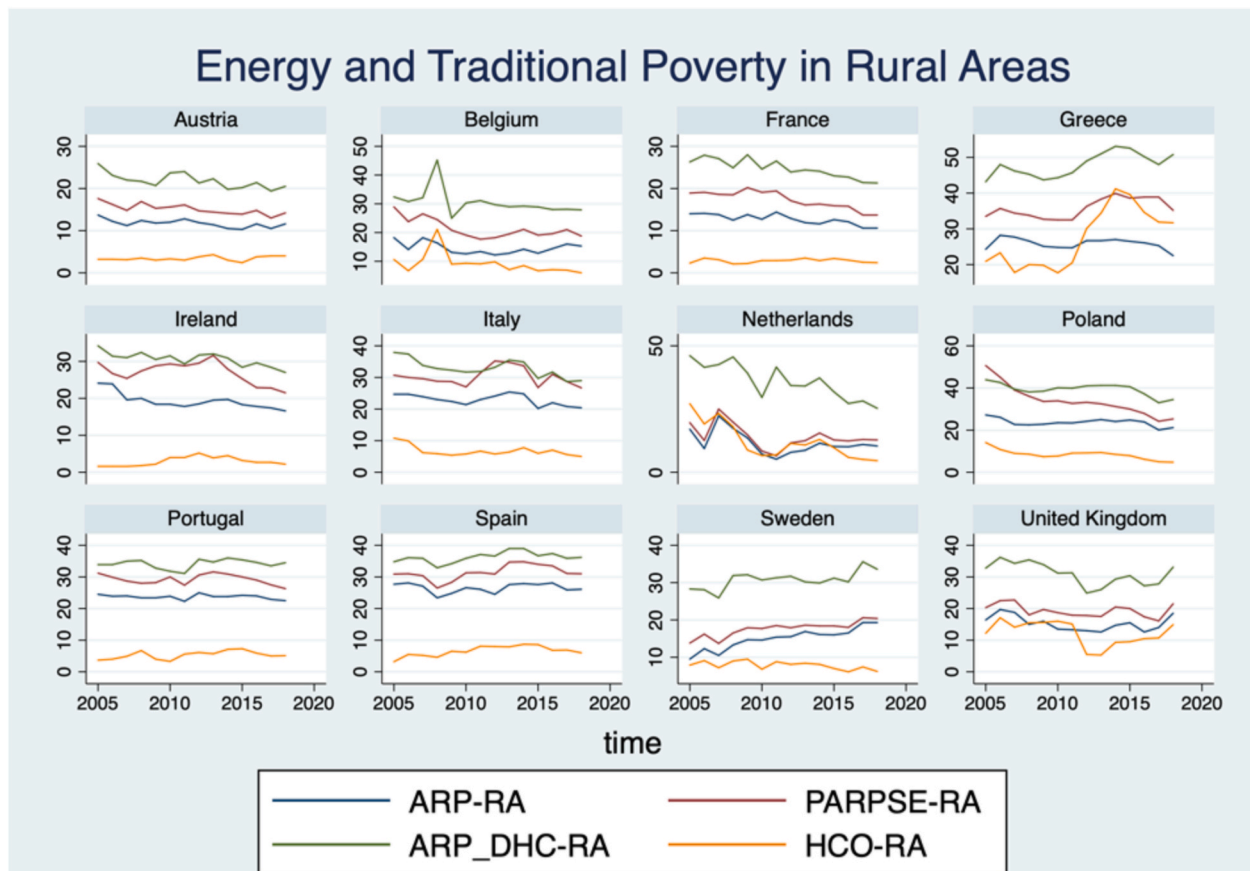


Fig. 1. (continued).

Table 3
Descriptive statistics of energy related variables.

Variable	Definition	Mean	SD.	Min	Max
Residential sector					
OIL	Oil consumption	2.035735	1.925819	.0358	9.646
GAS	Gas consumption	6.345861	7.923479	.0302	30.1493
WOOD	Wood consumption	2.168101	2.369035	.0198	9.1178
ELEC	Electricity consumption	4.10478	3.7455	.646	14.0243
Residential sector electricity consumption					
SPACE	for space heating	.7237923	.9551975	.0735	3.997
WATER	for water heating	.4237708	.503305	.0245	2.1556
COOK	for cooking	.305431	.2561703	.0653	.9946
APP_LIGH	for electrical appliances and lighting	2.628878	2.298794	.2954	7.4597

Notes: The number of observations is 168, and all variables are expressed in annual million tonnes of oil equivalent.

displayed in Fig. 3. The results of the analysis of the impact of forms of residential consumption, and of residential electricity consumption by end-use, on traditional poverty, by each degree of urbanisation are reported in Tables 9 and 10, respectively. Table S2, in the supplementary material, shows the average consumption of households by energy form, in each European country.

Overall, the results show that different forms of residential energy consumption have dissimilar impacts, and vary between areas with differing degrees of urbanisation. This reveals that the households in

areas with different levels of urbanisation attribute different uses to each form of residential energy consumption. Some forms of energy consumption lower their risk of experiencing energy poverty, while others increase it. For instance, Table 7 shows that electricity consumption increases a household's risk of energy poverty in rural areas, while natural gas and wood/biomass decrease the percentage of households suffering from energy poverty. It should be noted that wood/biomass has the ability to alleviate energy poverty in all degrees of urbanisation but with different magnitudes of impacts. Conversely, residential oil consumption increases the percentage of households threatened by energy poverty for every degree of urbanisation.

Table 8 and Fig. 3 show the impact of residential electricity consumption in detail, by exploring its impact by end-use activity. For instance, Fig. 2 shows that residential end-use consumption in towns and suburbs increases the risk of energy poverty. Fig. 3, takes this further, and shows that this negative impact is mainly caused by electricity consumption by appliances, lighting, and water heating. In fact, in towns and suburbs, electricity consumption for water heating and household appliances and lighting tends to increase the *at-risk-of-poverty rate after deducting housing costs* (see Table 8). At the same time, electricity consumption for cooking and space heating tends to reduce the risk of poverty. In rural areas, the increased risk of energy poverty due to residential electricity consumption (shown by both indicators used) is also mainly due to electricity consumption for water heating, household appliances, and lighting. In rural areas, residential electricity consumption for cooking and for space heating seems to reduce the housing-costs overburden. Nonetheless, if these end use activities are powered by electricity, they do not help reduce a household's risk of energy poverty in rural areas.

Electricity is considered the most promising form of green energy to achieve a low carbon economy because of its potential exploitation of

Table 4
Second generation unit root tests.

	Level		First differences	
	Without trend	With trend	Without trend	With trend
LARP-C	-1.9**	-0.74	-5.838***	-4.101***
LARP-TS	-1.73**	-1.96**	-7.136***	-4.781***
LARP-RA	-0.89	-1.27	-7.594***	-5.565***
LARPSE-C	-1.81**	-1.14	-5.602***	-5.07***
LARPSE-TS	-0.6	-2.45***	-5.783***	-3.551***
LARPSE-RA	0.05	-0.2	-5.895***	-4.033***
LARP_DHC-C	-4.76***	-2.11**	-6.161***	-4.461***
LARP_DHC-TS	-2.42***	-3.52***	-7.618***	-4.812***
LARP_DHC-RA	-3.48***	-3.29***	-8.052***	-5.7***
LHCO-C	-2.56***	-0.39	-4.772***	-3.736***
LHCO-TS	-3.79***	-4.66***	-8.014***	-6***
LHCO-RA	-1.55*	-1.22	-6.44***	-4.646***
LRPP-C	-0.32	0.67	-4.494***	-3.126***
LRPP-TS	-3.08***	-0.94	-5.697***	-3.913***
LRPP-RA	-1.56*	-2.74***	-7.201***	-5.807***
LINC-C	-0.33	1.51	-3.842***	-2.007**
LINC-TS	-0.11	0.87	-5.469***	-4.248***
LINC-RA	-0.26	-0.04	-5.723***	-4.612***
LHLWI-C	-1.68**	-0.51	-5.793***	-6.065***
LHLWI-TS	-1.15	-0.5	-5.416***	-3.985***
LHLWI-RA	-1.95**	-0.42	-5.303***	-4.22***
LOIL	-2.26**	0.09	-3.585***	-2.108**
LGAS	-2.14**	-1.19	-4.956***	-3.061***
LWOOD	-0.75	-0.95	-4.481***	-2.243**
LELEC	0.06	-2.02**	-5.298***	-3.065***
LSPACE	-0.92	0.76	-4.262***	-3.28***
LWATER	-0.16	0.84	-3.343***	-2.728***
LCOOK	-0.9	0.9	-5.122***	-4.535***
LAPP_LIGH	0.89	-0.44	-5.024***	-3.4***

Notes: *, **, *** denotes the statistical significance at the 10%, 5%, and 1% level, respectively. The null hypothesis of the CIPS test is the series integrated of order one.

natural resources such as wind and sun. In addition, electricity is the form of energy most consumed by European countries (see Table S2 in the supplementary material). Nevertheless, the energy transition and the electrification of economies it entails, may put even more pressure on households already threatened by energy poverty. The results of this research highlight that electricity only played a crucial role in

decreasing energy poverty in cities, and even there, residential electricity consumption for water heating, household appliances and lighting still increased the risk of energy poverty (see Table 8 and Fig. 3). These results reveal that energy poverty may be more related to infrastructure (stock) and energy-related education (stock) than to low-income (flow). In view of this, energy efficiency measures, and time-of-use tariffs, could alleviate energy poverty issues in cities (policies further discussed in the next section).

Wood/biomass is often used for space heating and cooking, and can also be used for water heating. Table 8 shows that this form of residential energy consumption reduces the risk of energy poverty whatever the degree of urbanisation. However, the use of this energy form is linked with increasing traditional poverty in both cities, and towns and suburbs. These dissimilar outcomes could be due to the limited availability of wood/biomass in more urbanised areas, in contrast to rural areas, where it is not subject to the same availability issues or costs for transportation and storage, and reduces both energy and traditional poverty. Furthermore, the availability of biomass in rural areas may even generate business opportunities, leading to more social cohesion and environmental protection.

It seems that residential oil consumption, mainly used to power old space heating equipment, tends to increase energy poverty (see Table 7 and Fig. 2). Nevertheless, it should be pointed out that, in cities and rural areas, this form of energy consumption apparently reduces poverty (see Table 9). This could be explained by the fact that bigger oil-fired boilers can be shared between households in large residential buildings, allowing them to adequately heat their homes more cheaply and thereby reduce traditional poverty. However, if the installation/equipment is old or poorly maintained, families may still be unable to maintain an adequate temperature in their rooms, or avoid rising costs, increasing both energy and traditional poverty. This research reveals that using electricity for space heating leads to a decrease in energy poverty and traditional poverty, so that the replacement of inefficient old oil-fuelled systems with electrical systems might represent a solution for large buildings.

The results of analysing energy poverty and traditional poverty with respect to the end-use of residential electricity consumption, suggest ways of alleviating the poverty. For instance, using electricity for cooking could reduce both energy and traditional poverty in cities,

Table 5
Models' nomenclature.

		Explained variable	Degree of urbanisation	Models' nomenclature
Forms of energy models	Traditional Poverty	At-risk-of-poverty	Cities	FEP-ATP-C
			Towns and suburbs	FEP-ATP-TS
			rural areas	FEP-ATP-RA
		People at risk of poverty or social exclusion	Cities	FEP-PPSE-C
			Towns and suburbs	FEP-PPSE-TS
			Rural areas	FEP-PPSE-RA
	Energy poverty models	At-risk-of-poverty after deducting housing costs	Cities	FEPP-ATP-C
			Towns and suburbs	FEPP-ATP-TS
			Rural areas	FEPP-ATP-RA
		Housing cost overburden	Cities	FEPP-HCO-C
			Towns and suburbs	FEPP-HCO-TS
			Rural areas	FEPP-HCO-RA
End-use electricity models	Traditional Poverty	At-risk-of-poverty	Cities	EEP-ATP-C
			Towns and suburbs	EEP-ATP-TS
			rural areas	EEP-ATP-RA
		People at risk of poverty or social exclusion	Cities	EEP-PPSE-C
			Towns and suburbs	EEP-PPSE-TS
			Rural areas	EEP-PPSE-RA
	Energy poverty models	At-risk-of-poverty after deducting housing costs	Cities	EEEP-ATP-C
			Towns and suburbs	EEEP-ATP-TS
			Rural areas	EEEP-ATP-RA
		Housing cost overburden	Cities	EEEP-HCO-C
			Towns and suburbs	EEEP-HCO-TS
			Rural areas	EEEP-HCO-RA

Table 6

Models specification tests.

Models			RESET	Modified Wald	Wooldridge	Breusch-Pagan LM	Pesaran	Freese	Friedman	Hausman RE vs. FE		
Forms of energy models	Traditional Poverty	FEP-ATP-C	2.08	109.92***	1.401	66.23	0.59	0.01	16.36	2.49		
		FEP-ATP-TS	1.92	124.94***	0.201	96.57***	1.09	0.38**	16.92	1.71		
		FEP-ATP-RA	2.58	2323.38***	13.43***	70.37	−0.54	0.13	9.94	1.66		
		FEP-PPSE-C	1.99	17.67	0.45	84.71*	4.15***	−0.02	28.21***	9.82		
		FEP-PPSE-TS	1.54	32.80***	0.46	75.90	2.06**	−0.01	17.52*	3.67		
		FEP-PPSE-RA	1.84	2142.08***	1.86	71.98	−0.67	−0.01	9.03	2.68		
	Energy poverty	FEFP-ATP-C	2.34	128.37***	1.18	69.08	0.51	0.04	16.18	0.41		
		FEFP-ATP-TS	2.04	57.57***	1.18	95.69***	2.19***	0.22*	21.61**	0.72		
		FEFP-ATP-RA	0.52	160.20***	0.15	74.82	3.49***	0.05	25.61***	1.83		
		FEFP-HCO-C	1.41	109.15***	2.56	79.76	−0.77	−0.01	16.09	0.7		
		FEFP-HCO-TS	0.87	30.39***	0.19	61.56	−0.99	−0.01	10.26	0.38		
		FEFP-HCO-RA	1.45	54.23***	1.55	64.8	0.68	−0.18	15.17	1.81		
		End-use electricity models	Traditional Poverty	EEP-ATP-C	0.72	131.99***	1.3	63.9	0.58	−0.05	15.09	2.52
				EEP-ATP-TS	1.88	91.02***	0.04	92.37**	1.34	0.41	16.51	0.39
EEP-ATP-RA	1.26			2255.1***	1.53	79.72	−0.66	0.14	8.35	0.85		
EEP-PPSE-C	0.97			17.13	0.55	84.09*	4.37***	−0.02	28.32***	10.88		
EEP-PPSE-TS	1.88			34.87***	0.01	69.34	1.71*	0.03	17.52*	2.58		
EEP-PPSE-RA	1.96			2432.73***	0.19	70.66	−0.28	−0.08	9.97	1.93		
Energy poverty	EEEP-ATP-C		1.25	155.07***	0.35	68.58	0.47	0.12	15.92	1.02		
	EEEP-ATP-TS		1.35	45.32***	0.65	101.46***	2.43**	0.25*	20.15**	0.79		
	EEEP-ATP-RA		0.81	258.54***	0.57	69.09	3.67***	0	27.66***	2.05		
	EEEP-HCO-C		1.79	102.66***	1.89	74.79	−0.31	−0.03	15.47	1.25		
	EEEP-HCO-TS	0.97	30.1***	0.79	64.98	−0.25	0.04	9.92	1.82			
	EEEP-HCO-RA	0.46	46.53***	1.49	63.89	2.06**	−0.17	17.01	3.17			

Notes: ***, **, * denote statistical significance at the 1%, 5%, and 10% level, respectively. the Ramsey RESET tests H0: the model is well-fitted, with no omitted variables; the modified Wald test has χ^2 distribution and tests H0: $\sigma_c^2 = \sigma^2$, for $c = 1, \dots, N$; the Wooldridge test is normally distributed $N(0,1)$, and tests H0: no serial correlation; Pesaran, Freese and Friedman test the H0: residuals are not correlated; Hausman results for H0: difference in coefficient of FE and RE is not systematic including the constant.

Table 7

Energy forms impact on energy poverty.

Models	FEFP-ATP-C	FEFP-HCO-C	FEFP-ATP-TS	FEFP-HCO-TS	FEFP-ATP-RA	FEFP-HCO-RA
Variables						
$\Delta LINC$	−0.2705***	−0.4805***	−0.4711***	−0.7292***	−0.7876***	−1.7089***
$\Delta LHLWI$	0.0839***	0.1298***	−0.0218***	−0.0308	−0.0438***	0.0727***
$\Delta LRPP$	−0.23***	−0.6386***	−0.1486***	0.0485	−0.5971***	−0.4242**
$\Delta LOIL$	0.0391*	0.0639	0.0278***	0.1900***	0.0858***	0.0559
$\Delta LGAS$	0.0186	−0.0198	−0.0649***	−0.1824***	−0.0734***	−0.0717
$\Delta LWOOD$	−0.0204*	−0.051	−0.0401***	−0.1035*	0.0124	−0.2060***
$\Delta LELEC$	−0.1903***	−0.5858***	0.0394**	−0.4477**	0.0553*	0.5716***
Trend	−0.0031***	−0.0077***		−0.0078***		−0.0114***
Constant	0.0316***	0.0658***	0.0122***	0.0889***	0.0164***	0.1089***

Notes: ***, **, * denote statistical significance at the 1%, 5%, and 10% level, respectively.

where electric cooking technologies are already widespread for new and renovated buildings. However, it would not yet have any noticeable effect in rural areas, or towns and suburbs, as areas with this level of urbanisation still lack of access to these new technologies. The greatest concern at every level of urbanisation is residential electrical water heating, as the study's results stress that electricity consumption for water heating has the greatest impact on increasing the percentage of households suffering from both energy poverty, and traditional poverty. Once again, this reinforces the idea that energy poverty is more a consequence of lack of stock rather than a consequence of flow (or income).

5. Discussion

This research highlights the importance of studying the impact of different forms of residential energy consumption on both energy and traditional poverty by degree of urbanisation. It shows that, depending on the degree of urbanisation, certain forms of residential energy

consumption can be more useful and improve the well-being of households and it confirms that energy poverty is dependent on the type of energy consumed and the level of urbanisation. Therefore, energy policies should not be planned uniformly on a European or national scale. Policy guidelines or targets, such as the share of renewables in the energy mix, and energy poverty reduction targets, should be devised at a suitably local level. The accessibility and prices of different forms of energy differ widely between countries, and within countries, by region, and degree of urbanisation. For example, the accessibility of energy in rural areas of southern Europe is quite different from that in northern Europe. Consequently, energy policies and measures should be planned at a more disaggregated level of sub-national regions, whose households have divergent energy needs, and varying access to forms of energy. Furthermore, the results of this research also confirm that no single form of energy consumption can reduce energy poverty for every degree of urbanisation. These results clearly show that the energy transition, and the higher residential consumption of electricity it entails, requires well-informed policy. This research also went one step further and analysed

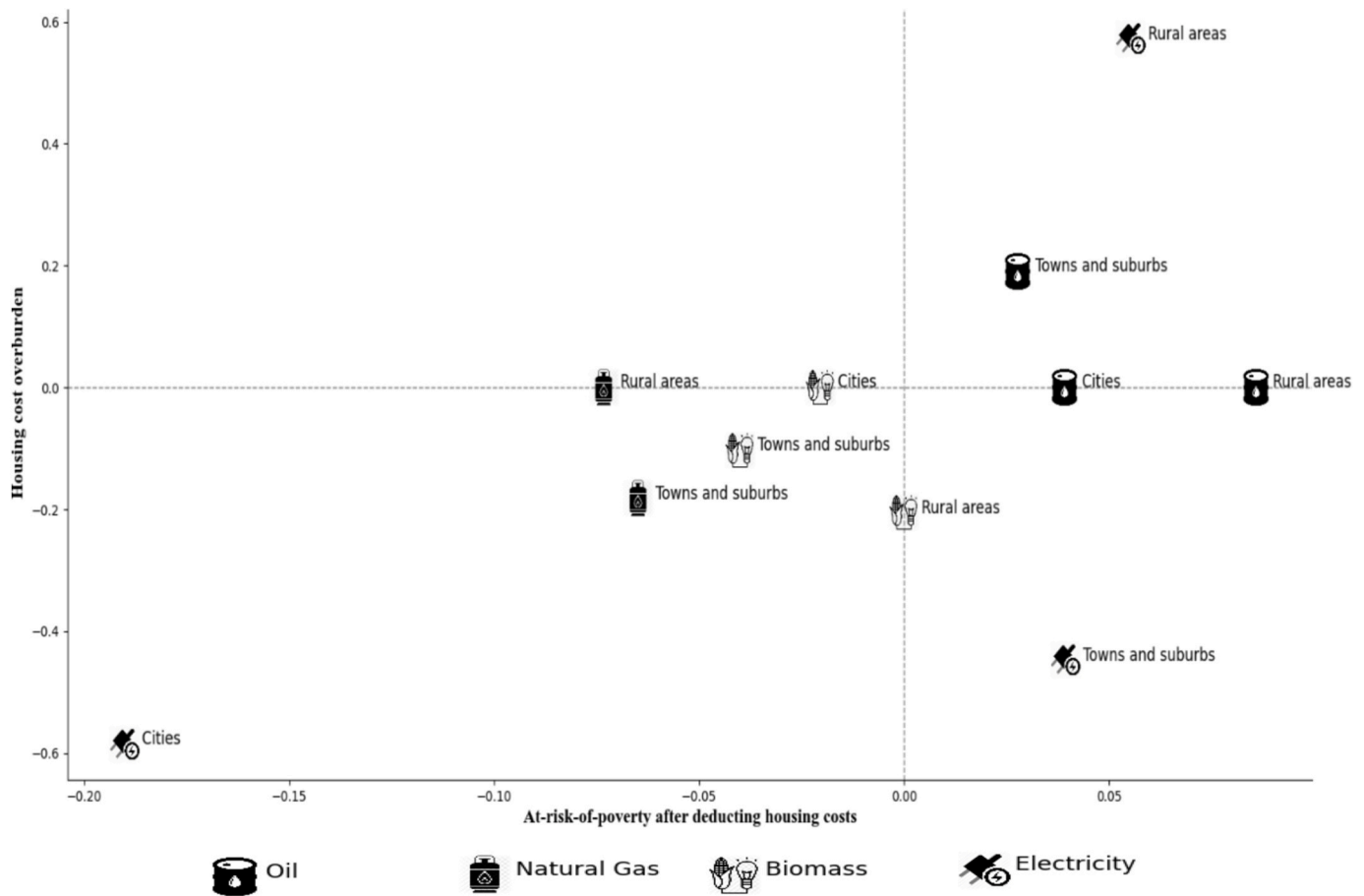


Fig. 2. Energy forms impact on energy poverty.

Table 8

End-use electricity consumption impact on energy poverty.

Models	EEEP-ATP-C	EEEP-HCO-C	EEEP-ATP-TS	EEEP-HCO-TS	EEEP-ATP-RA	EEEP-HCO-RA
Variables						
$\Delta LINC$	-0.2480***	-0.4576***	-0.5030***	-0.8182***	-0.7523***	-1.6487***
$\Delta LHLWI$	0.1149***	0.1519***	-0.0353***	-0.0297	-0.0444***	0.1012***
$\Delta LRPP$	-0.2019***	-0.6715***	-0.1637***	-0.0864	-0.4629***	-0.1400***
$\Delta LSPACE$	-0.0612***	-0.1530***	-0.0586***	-0.2363***	-0.0006	-0.2135***
$\Delta LWATER$	0.0733***	0.1523***	0.0615***	0.3604***	0.0709**	0.5057***
$\Delta LCOOK$	-0.2048***	-0.2412***	-0.0698***	-0.0889	0.0153	-0.6409***
$\Delta LAPP_LIGH$	0.1120***	-0.2433*	0.0866**	-0.5842***	0.0636	0.5073***
Trend		-0.0060***				
Constant	0.0094***	0.0499***	0.0144***	0.0269***	0.0102***	0.0183***

Notes: ***, **, * denote statistical significance at 1%, 5%, and 10% level, respectively.

the impact on poverty of residential electricity consumption by end-use, and provided important knowledge for more accurately targeted policy planning to assist households during the coming energy transition. The energy transition can provide benefits for households, but governments need to tailor policies to match the varying needs and circumstances of households.

Some forms of residential energy consumption can provide households or individuals with better living standards, while others can have the reverse impact, as this research shows. For instance, in rural areas, wood/biomass is an endogenous and mostly freely available energy form. The consumption of this energy form can be used for basic households needs, such as water heating, space heating, and cooking. It can also provide greater well-being for families because its low or nearly zero costs reduces energy expenses. It should also be noted that the consumption of wood/biomass can also substitute the use of gas.

Consequently, the higher the consumption of wood/biomass in rural areas, the more of the household budget remains available for other primary needs. This means that wood/biomass can reduce the risk of energy poverty, traditional poverty and social exclusion. In contrast to rural areas, the results show that in cities, and in towns and suburbs, the consumption of wood/biomass has cost limitations. In these areas with greater levels of urbanisation, wood is not freely available and may involve transport costs and storage restrictions. This is confirmed by the fact that wood/biomass was found to increase energy poverty in more urbanised areas such as cities, towns and suburbs.

Gas consumption constituted almost half of total household energy consumption (see Table 3). The results show that this form of energy was generally preferred by households in rural areas, and in towns and suburbs, making natural gas is a key form of energy for reducing energy poverty in these less urbanised areas. When we turn to cities, the results

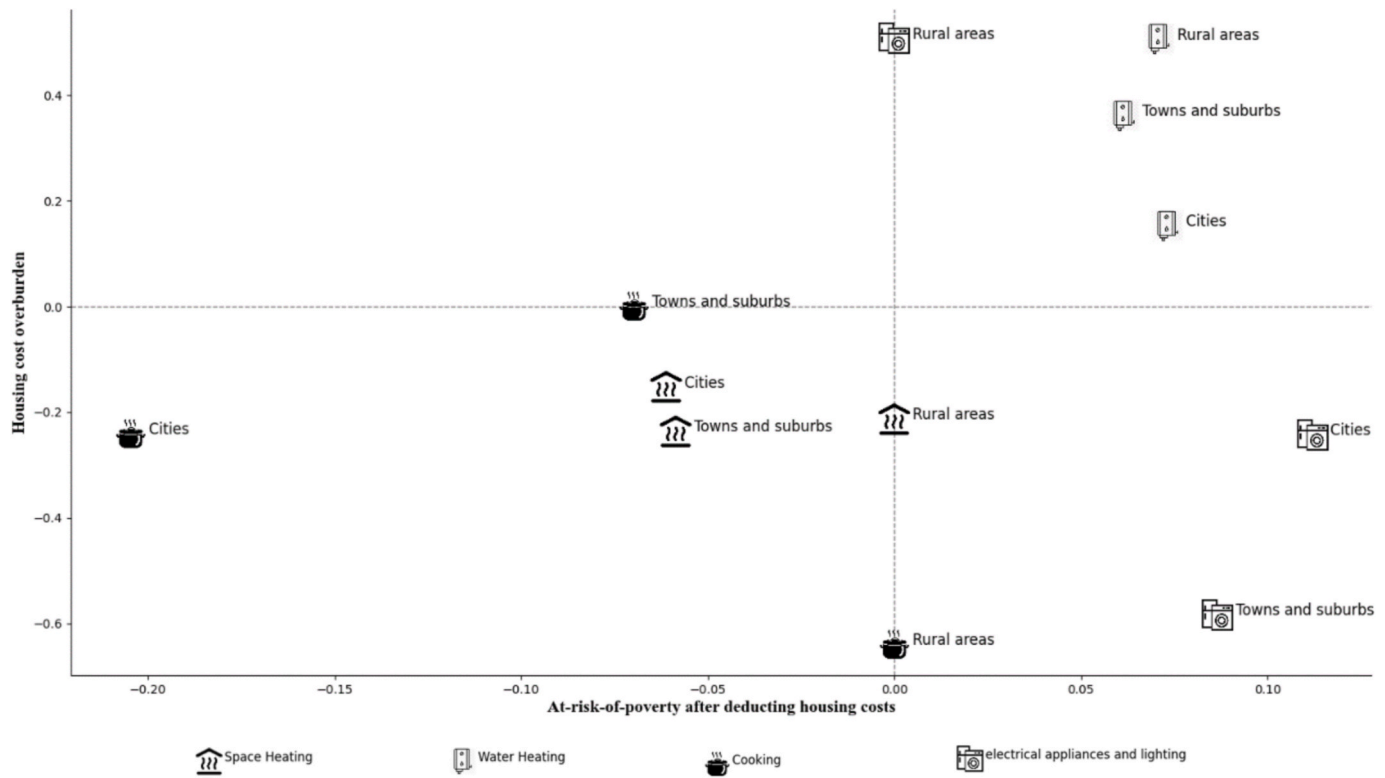


Fig. 3. End-use electricity consumption impact on energy poverty.

Table 9
Energy forms impact on poverty.

Models	FEP-ATP-C	FEP-PPSE-C	FEP-ATP-TS	FEP-PPSE-TS	FEP-ATP-RA	FEP-PPSE-RA
Variables						
$\Delta LINC$	0.0219	-0.2184***	-0.5141***	-0.7193***	-0.4780***	-0.4817***
$\Delta LHLWI$	0.1677***	0.1901***	-0.0589***	0.0368***	0.0455***	0.1299***
$\Delta LRPP$	-0.3118***	-0.2463***	-0.1392**	-0.074	-0.6220***	-0.3594***
$\Delta LOIL$	-0.0491***	-0.0166**	0.0013	0.0655***	-0.0533***	-0.0260**
$\Delta LGAS$	0.0538***	0.0352***	-0.1284***	-0.1050***	0.0052	-0.0336**
$\Delta LWOOD$	0.0353**	0.0778***	0.01	0.0506***	-0.2418***	-0.2121***
$\Delta LELEC$	-0.0881*	-0.1545***	0.1952***	0.1821***	0.2855***	0.2249***
Trend				-0.0030***		
Constant	0.0038	0.0029	0.0170***	0.0313***	0.0109***	0.0022

Notes: ***, **, * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 10
End-use electricity consumption impact on poverty.

Models	EEP-ATP-C	EEP-PPSE-C	EEP-ATP-TS	EEP-PPSE-TS	EEP-ATP-RA	EEP-PPSE-RA
Variables						
$\Delta LINC$	0.0133	-0.3332***	-0.5052***	-0.7359***	-0.4462***	-0.3829***
$\Delta LHLWI$	0.1728***	0.1617***	-0.0793***	0.0275***	0.0165	0.1312***
$\Delta LRPP$	-0.2450***	-0.2370***	-0.2153***	-0.0769*	-0.5216***	-0.3911***
$\Delta LSPACE$	-0.0269*	0.0279*	0.0303**	0.0931***	-0.1860***	-0.1122***
$\Delta LWATER$	-0.0095	-0.0216	0.0631***	-0.011	0.2108***	0.1883***
$\Delta LCOOK$	-0.1144***	0.0507	0.1443***	0.1241***	-0.1498***	-0.0311
$\Delta LAPP_LIGH$	0.0763***	-0.1620***	-0.1915***	-0.0513*	0.1931**	0.0848*
Trend		-0.0030***		-0.0039***		
Constant	0.0122***	0.0285***	0.0175***	0.0395***	0.0039	-0.0055***

Notes: ***, **, * denote statistical significance at the 1%, 5%, and 10% level, respectively.

show that natural gas has no impact on the indicators for energy poverty, but tends to increase the traditional poverty indicators. This means that natural gas has been increasing traditional poverty in city households. The high dependence of certain households on natural gas is

also a major vulnerability. The unforeseen recent escalation in natural gas prices due to geopolitical issues, cuts in financial investment, and internal and external dependence, has led European countries to switch to a strategy of conserving natural gas. This strategy of conservation may

cause abrupt rises in natural gas prices and thus in electricity prices, causing those struggling with energy costs to fall into energy poverty. Consequently, it is essential to encourage households to switch from primary energy sources to electricity, an energy source that can be generated using indigenous green energy sources.

The results show that electricity consumption increases energy poverty and traditional poverty in both rural areas, and towns and suburbs, whereas, in cities, electricity consumption can reduce energy poverty. This divergence arises because city lifestyles are more compatible with electricity consumption than those of households in non-urban areas. In cities, there is widespread access to modern and energy-efficient technologies powered by electricity which can increase the well-being of families and reduce their energy and traditional poverty. For instance, the use of modern electric cooking technologies has decreased energy poverty in cities. Electricity can power household appliances and devices, such as televisions, phones, laptops, and other gadgets, thereby improving individual well-being. These devices can be used to access entertainment and learning processes for adults and children, potentially increasing their present and future productivity and income. In other words, electricity consumption could reduce their propensity to suffer from both energy and traditional poverty. However, the results of our research also show that electricity consumption for certain kinds of end use can increase energy poverty, suggesting that the impact of electricity consumption on energy poverty may be more related to energy-related infrastructure and education (stock) than to low income (flow).

This research also found that the biggest driver of energy poverty was the use of inefficient technologies in water heating systems, and household appliances, such as washing machines, lighting, and entertainment. Inefficient technologies can cause high electricity consumption while reducing their usefulness and the well-being of families. In contrast, highly energy-efficient technologies consume less electricity and are at least as useful and beneficial to households as less efficient technologies. Consequently, policymakers should consider subsidies for the rehabilitation of energy-inefficient dwellings and appliances. This funding would benefit households through the potential savings of energy efficiency. Rehabilitation measures should be introduced for single-family and large residential buildings. These measures could be modest, for instance, painting the facades and roofs with thermochromic colours, or more extensive, such as applying better wall insulation, installing thermally efficient glazing, and exchanging household appliances for more energy-efficient ones, or those with deferred start to take advantage of time-of-use pricing strategies. This cost-effective funding of energy-efficiency improvements for households would enable a long-term and sustainable energy poverty solution.

The decentralisation of electricity production is another potential solution for reducing the threat energy poverty represents to households in rural areas, cities, and suburbs during the energy transition. In other words, the integration of decentralised electricity generation from renewable sources could be a step towards overcoming both energy and traditional poverty. Additionally, this decentralisation could also be a way of encouraging consumers in rural areas to switch from primary energy sources to electricity, by providing them with 'free' electricity generation. Solar photovoltaic (PV) panels can be installed on the roofs of houses or buildings make it possible to produce electricity locally. The main barrier to this solution is the initial investment. Vulnerable households, such as those on low incomes or those threatened by energy poverty, cannot afford the cost of solar PV panels. Policymakers should provide funding schemes to increase the penetration of solar PV panels in single or multi-family residential buildings. The capacity of solar PV panels should be sufficient to satisfy residential-based consumption, i.e., the continuous consumption of electricity during the day to power refrigerators, freezers, and other appliances. This will contribute to overcoming energy poverty and could even provide savings to overcome traditional poverty issues.

In summary, the results of this research show that energy policies

should not be uniformly applied at a national level. The impact of forms of residential consumption and the impact of residential electricity consumption by end-use vary between areas with differing degrees of urbanisation. Consequently, energy policies aimed at alleviating poverty, and encouraging the energy transition and consequent electrification of economies, should be planned at a regional level, or by level of urbanisation. This planning could also take advantage of the strengths of local forms of clean energy. Areas with high solar exposure could promote the installation of both solar PV panels to generate electricity, and solar thermal collectors, to reduce electricity consumption for water heating systems. It should also be emphasised that, in rural areas, towns, and suburbs, strategies for reducing the reliance on wood/biomass and natural gas can trigger effects on both poverty and energy poverty. Therefore, the energy transition and the consequent electrification of economies should be carefully planned so as not to worsen both poverty and energy poverty in these areas.

6. Conclusion and policy implications

This research used the Eurostat database to perform an analysis of the impacts of forms of residential energy consumption and residential electricity consumption by end-use on energy poverty and energy poverty. Traditional poverty and energy poverty in twelve EU countries were assessed for a period from 2005 to 2018 by degree of urbanisation. A battery of tests were conducted to evaluate the characteristics of the panel data and the specification of the models. The tests indicated that the FGLS estimator was best suited to handling the characteristics of the data and the specifications of the 24 models. The findings of this research provide a solid foundation from which decision-makers can gain insights on energy poverty policies.

Overall, the results prove that different forms of energy have differing effects on areas with different degrees of urbanisation. In other words, there is no single energy form for eradicating both poverty and energy poverty. Up until now, energy policies have generally been planned and executed at a national level. However, it would be better if these policies were planned within countries at a regional level, taking into account the local characteristics of residential consumption and its impact on energy poverty. In cities, residential electricity consumption was found to reduce energy poverty, while in rural areas, towns and suburbs, natural gas and wood/biomass were the crucial forms of energy for reducing energy poverty. In fact, residential electricity consumption in these less-urbanised areas increases both poverty and energy poverty. To understand residential electricity consumption in more detail, it was further disaggregated by end-use. Water heating, household appliances and lighting were found to be the key elements increasing energy poverty in rural areas, towns and suburbs.

The findings of this research suggest that energy poverty is more due to a lack of knowledge and financial investment than a consequence of low-income. Energy policies need to evolve to protect households threatened by energy poverty during the ongoing energy transition and electrification of the economy. Therefore, governments ought to be prepared to support public investment through: (i) the provision of energy-efficient appliances; (ii) the installation of solar PV panels; (iii) the installation of solar thermal collectors; (iv) a combination of both (ii) and (iii) with energy storage options and incorporated in energy communities; (v) education campaigns about the use of technologies in (ii), (iii), and (iv); (vi) information campaigns and training about price-responsive tariffs and demand response programmes; and (vii) the rehabilitation of low energy-efficient buildings, particularly those occupied by families threatened by energy poverty.

Further research is needed to address some of the limitations encountered in this study, primarily the limited number of countries under analysis and the short time span studied. To overcome these obstacles, a single energy poverty measure should be defined by the EU. A data set on energy poverty could then be set up, making it easier to compare the drivers of energy poverty in EU countries and their regions.

It should also be noted that microeconomic data sets, such as the EU-SILC, based on surveys of families could also produce more useful statistics about this topic. As the local circumstances of households have been shown to be so important, future EU inquiries about energy poverty should include questions about the forms of energy they consume and their expected costs, as well as the consumer's preferences. Lastly, this study was unable to take into account the coping strategies that some households perform to combat energy poverty (such as only heating one room). Further research is needed to study both these coping strategies and, to measure other forms of hidden energy poverty.

CRedit authorship contribution statement

Diogo Santos Pereira: Conceptualization, Data Formal analysis, Investigation, Resources, Writing – original draft, Visualization.
António Cardoso Marques: Validation, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2022.113346>.

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